Wheelchair Tire Cleaning Station

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Client: Ryan & Oliver





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Problem Statement

Ryan frequently transitions between outdoor and indoor environments using his powered wheelchair, causing his tires to collect dirt, mud, snow, salt, and debris that are tracked into his home. Cleaning the tires can be difficult and may require caregiver assistance, reducing independence and adding inconvenience to daily routines. The goal of this project is to design an accessible, low-effort tire cleaning system that allows Ryan to independently clean his wheelchair tires when entering his home.

Meet Our Client: Ryan (and Oliver)

Ryan is an active and independent individual who uses a powered wheelchair following a C4 spinal cord injury.

He frequently moves between outdoor and indoor environments throughout the day.

Ryan values accessibility, efficiency, and maintaining independence in his daily routine.

He is also accompanied by his service dog, Oliver.



Concept Development

P1 – Brush Ramp
Ryan rolls forward and backward over brush rollers that remove dirt from the tires.



May require multiple passes

P2 – Wheel Cleaning Arm
A brush attached to the wheelchair frame contacts the wheel and cleans it as the chair moves.



Complex mounting, adds weight

P3 – Cleaning Platform
Ryan rolls onto a platform where brushes and rollers allow the wheels to spin, removing debris.



Complex mounting, adds weight

P4 – Reach Brush
A long curved brush allows Ryan to clean his wheels while seated in the wheelchair.



Requires arm reach and manual effort

User Needs

User Need	Design Requirement
Independent use	No caregiver assistance required
Effective cleaning	Removes dirt, mud, snow, debris
Low effort	Minimal force or dexterity needed
Fast use	Under 1 minute to operate
Safe indoors	Protects flooring / prevents slipping
Low maintenance	Easy to empty and clean
Compact size	Fits standard doorway/entryway

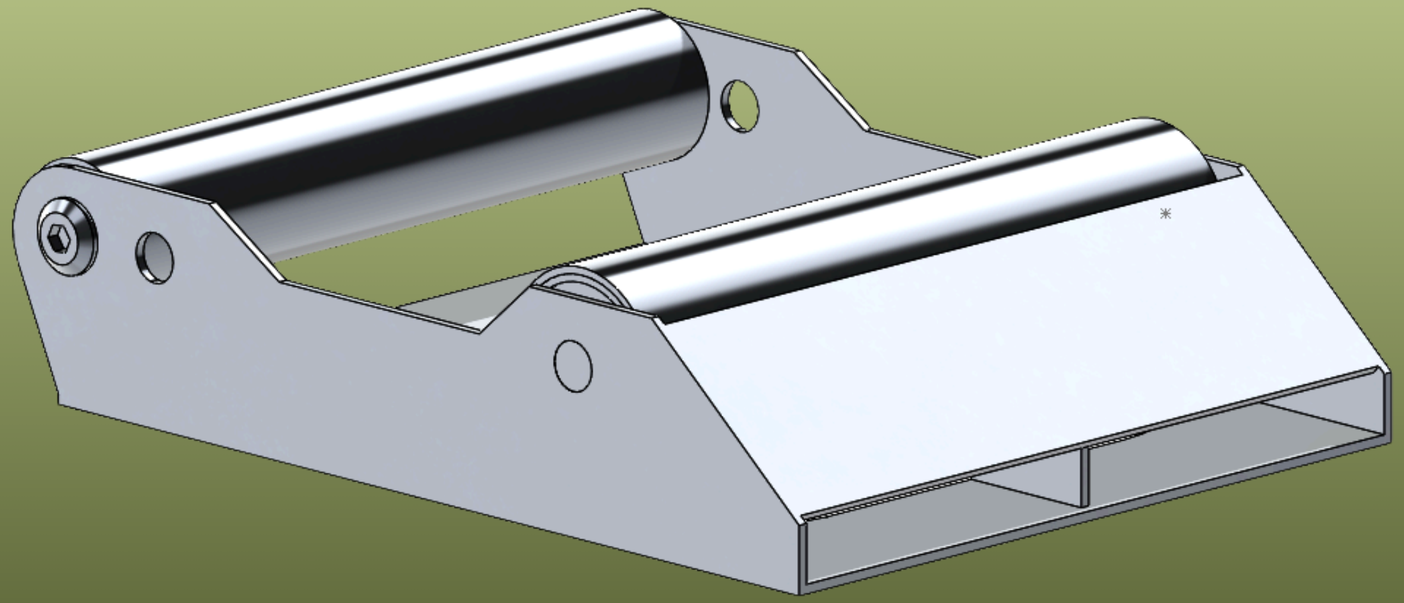
Engineering Requirements

Specification	Target Value
Cleaning Time	< 60 seconds
Device Width	≤ 18 cm
Device Length	≤ 30 cm
Wheel Load Capacity	≥ 350 lb total system load
Operating Method	Drive-on / no hand use required
Surface Safety	Non-slip base with stable frame
Maintenance Time	< 2 minutes to clean debris tray
Portability	Lightweight and movable by caregiver

Final Direction

P3 – The cleaning platform was selected as the best balance of independence, effectiveness, safety, and ease of use.

Ryan can clean wheelchair tires without hand contact or caregiver assistance.



Why we chose this design:

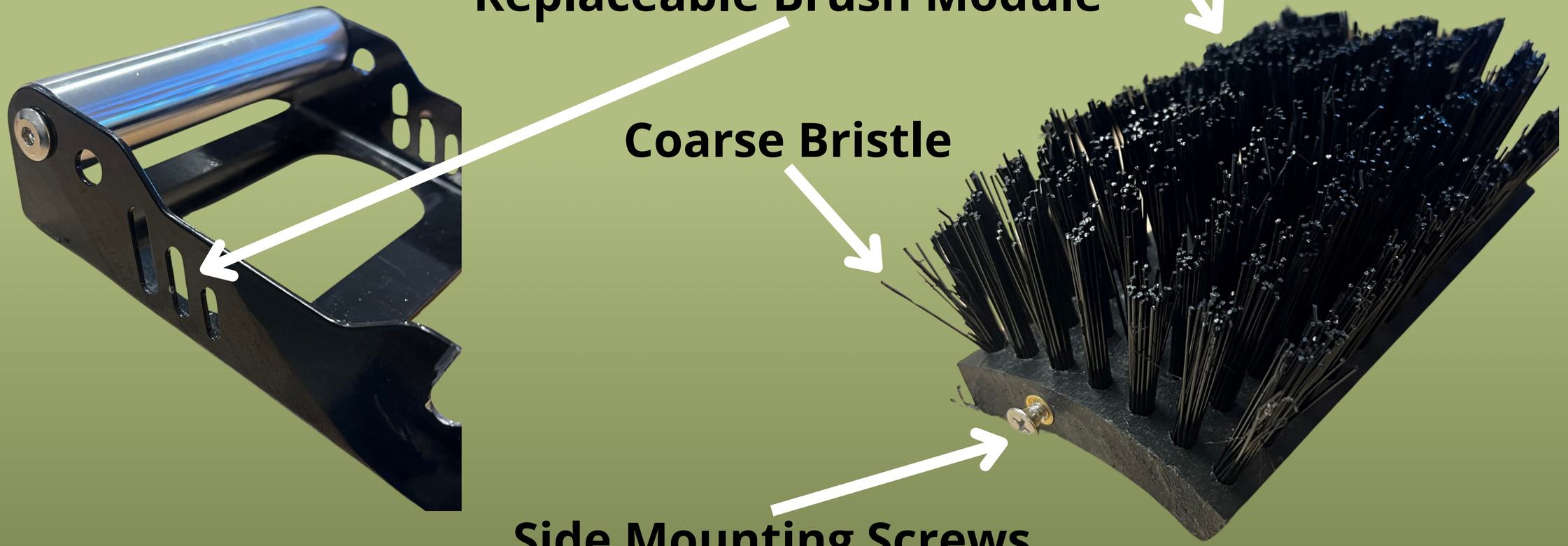
1. Independent operation
2. No hand use required
3. Strong cleaning potential
4. Stable use at doorway
5. Low daily effort
6. Easy to understand and use

Cleaning Brushes

Low-Cost Commercial Brush Material

Replaceable Brush Module

Coarse Bristle



Side Mounting Screws

BASE STRUCTURE



Starting Point

We used an off-the-shelf motorcycle wheel cleaning stand, originally designed to support wheels and allow free spinning during cleaning.

Why We Chose It

The base needed to be the strongest part of the system to support the wheelchair and user load. The stand was rated for 800, making it a strong starting platform.

Modifications

- Added custom one-way roller
- Modified structure for fit and stability
- Redesigned ramp for smoother entry
- Integrated useful existing features

Outcome

Using an existing heavy-duty base saved time while providing a durable and stable foundation.

Manufacturing Process

To bring the final concept to life, we used several fabrication methods:



Lathe Machining – custom one-way roller and axle features



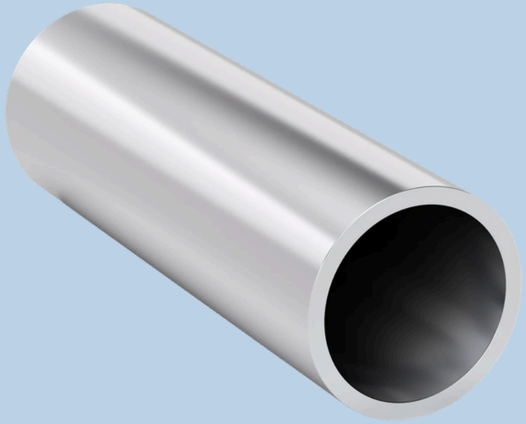
Waterjet Cutting – durable aluminum base and support geometry



Drill– Used to machine the keyway slot for the one-way roller assembly.

Custom One-Way Roller Development

No commercial one-way roller matched our required size and function, so we designed a custom version.







Construction

- **Outer Shell: Purchased aluminum tube**
- **Center Shaft: Aluminum shaft machined to fit bearing inner diameter**
- **One-Way Bearing: Press-fit into roller assembly**
- **Result: Roller spins one direction and locks in the other**

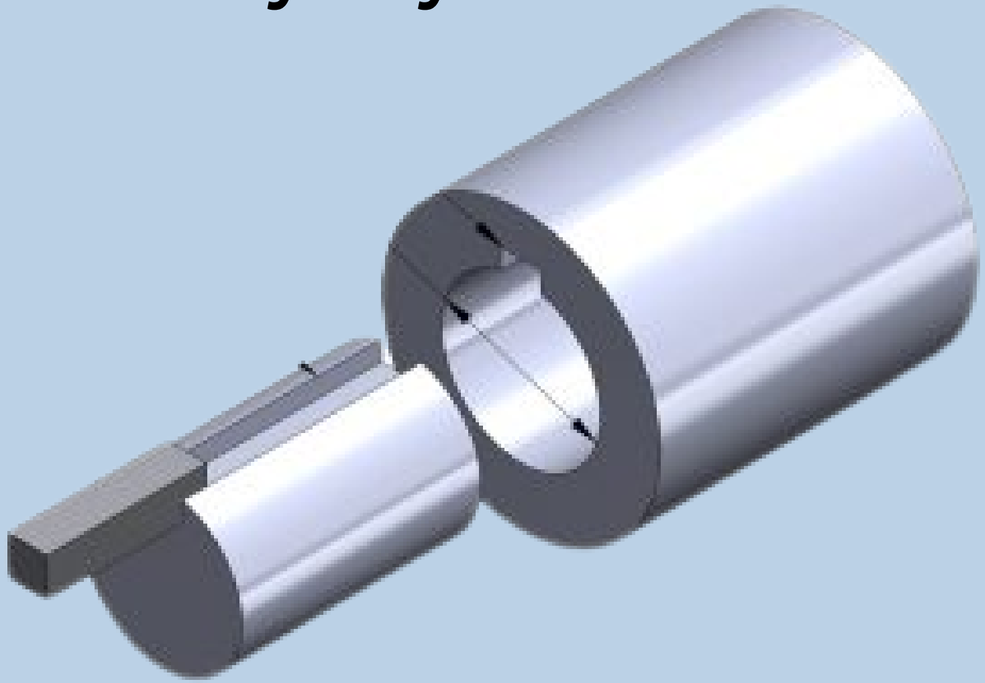
Assembly Method

All roller components were connected using a keyed shaft and keyway system. This method securely links the aluminum shaft to the surrounding roller assembly, allowing torque to transfer effectively between parts while preventing slipping during rotation. The keyway also improves alignment, durability, and reliability under repeated daily use.

Final Manufactured Version



Key-way Model



Failures







How the Cleaning Platform Works

1. Ryan approaches platform independently
2. Front wheels roll onto guide ramp
3. Tires contact brushes
4. Ryan drives his wheel forward making the wheelchair spin on itself
5. Dirt and debris are removed while passing through
6. Ryan exits platform into clean indoor space

User Testing Plan & Responses

Use Cases Considered

- Entry after rain or mud
- Entry after snow or salt exposure
- Daily repeated use
- Forward approach and wheel alignment
- Reverse exit maneuver
- Moving and storing the device between uses
- Cleaning and maintenance after debris buildup

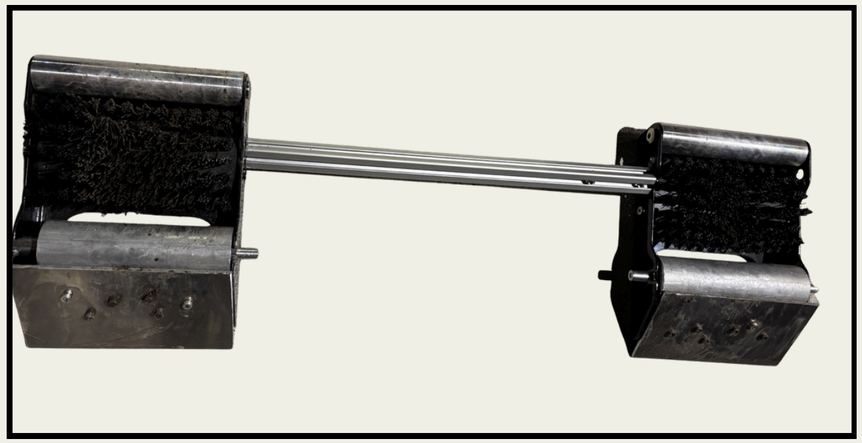
Key Responses

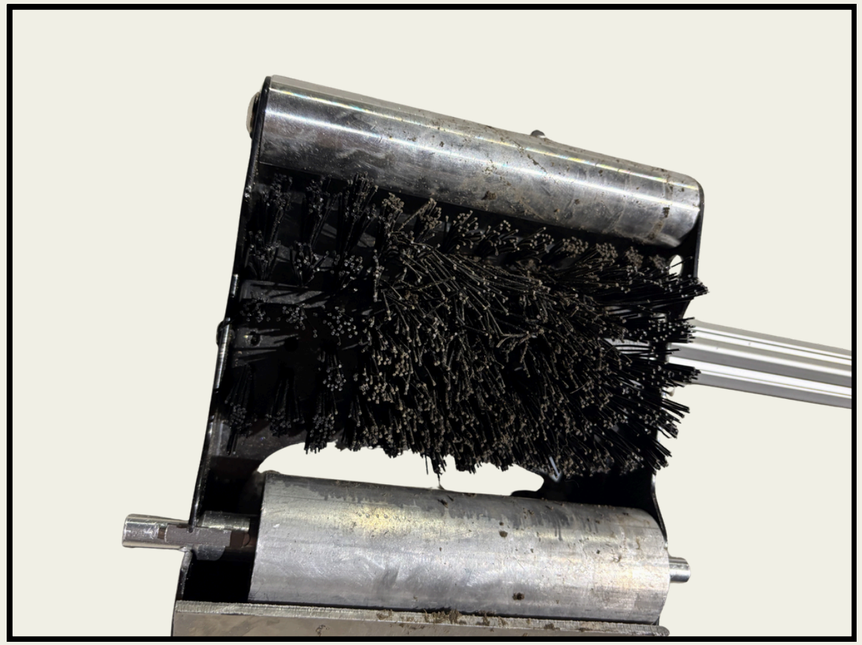
- Easy concept to understand
- Stable platform is essential
- Coarse brushes improve dirt removal from tire tread
- Removable brushes make cleaning/ replacement easier
- Lightweight design (under 10 lb) improves portability and storage compared to many ramps

Interpretation

Testing suggests that stability, effective cleaning performance, easy maintenance, and lightweight portability are the most important factors for successful long-term daily use.

Dry Run Demo





Demo with Mud

Before



After





took around 2 minutes to fully clean

Risk matrix

Scenario	Severity (1-5)	Likelihood (1-5)	Risk	Mitigations
Hand, foot, or clothing gets caught in rotating rollers	5 - Severe	2 - Unlikely	10	• Protective roller covers • Safe spacing without pinch points • Shielding around moving parts
User rolls too far forward and becomes destabilized or goes off the device	4 - Major	3 - Moderate	12	• Front wheel stopper • Visual or audible wheel position markers
User loses balance while entering or exiting device	5 - Severe	3 - Moderate	15	• Low incline ramp • High-friction surfaces • Side supports or rails
Device shifts or slides under wheelchair during use or reverse exit	5 - Severe	2 - Unlikely	10	• Anti-slip rubber base • Increased device weight or base dimensions • Anchoring to nearby stable structure
Mechanical failure, brush breaks, or roller stops	3 - Moderate	2 - Low	6	• Regular maintenance guidelines • Use of durable components

Lessons Learned

1. Simpler passive systems outperformed complex mounted concepts
2. Early prototypes helped reveal design flaws quickly
3. Independence mattered more than automation
4. User-centered iteration improved final decisions

Advice for Future Teams

Improve durability and long-term reliability by using wear-resistant roller and brush materials, while testing with full wheelchair dimensions and different tire sizes to improve compatibility across users.

Develop the design further for wet conditions by adding a sealed bottom tray or removable collection bin to capture water, mud, snow, and debris, with easier drainage and cleanup.

Explore foldable or lower-profile versions for easier storage near entryways, while continuing direct client feedback to keep the system practical, intuitive, and effective for daily use.

Design Validtion

User Need	Status
Independent use	✔ Achieved
Effective tire cleaning	✔ Achieved
Low effort operation	✔ Achieved
Safe and stable use	✔ Achieved
Compact	✔ Achieved
Easy maintenance	◐ In Progress

took around 2 minutes to fully clean

Deliverables to Client

- Presented a complete design concept developed to address the client’s accessibility challenge.
- Shared prototype progress and demonstrated how the system is intended to function.
- Provided design documentation, models, and engineering rationale for future teams.
- Identified recommended improvements and areas for further refinement before user testing.
- Established a strong foundation that can be further developed, validated, and eventually tested with the client.

Thank you!